L4.1 What is a basis for a vector space?

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online degree program

on data science. So, in this video, we are going to study what is the basis for a vector space.

So, just to recall what we have done as part of this module before, we have studied what

are vector spaces, we have seen the notion of a linearly dependent set in a vector space. And we

have seen the notion of a linearly independent set in a vector space. And, today's video

about a basis builds on the fact that we know what is linear independence of a set of vectors.

So, let us quickly recall this idea of linear dependence and linear independence.

So, suppose we have a set of n vectors, v1, v2, vn in a vector space V. so the set v1, v2, vn is said

to be linearly dependent, if there exists scalers a1, a2, an, not all 0, such that a1, v1 plus a2,

v2 plus all the way up to an vn is 0. So, this means there is a linear combination of Vi

which is a non-zero linear combination, which means that at least one of the coefficients is

non-zero. Note that this does not mean that all the coefficients are non-zero, but it means that

at least one of the coefficients is non-zero. So, we have a linear combination such that,

with non-zero coefficients and that linear combination is 0.

So, we are saying the question x1 v1 plus x2 v2 plus xn vn has a non-trivial solution,

this is what we saw last time. And now, linear independence is saying it is not

linearly dependent. So, explicitly what this means is that the set v1, v2, vn is said to be linearly

independent. If the only choice of scalars a1, a2, an such that the corresponding linear combination,

a1 v1 plus a2 v2 plus an vn is 0, so this equals 0 is with a1 is 0.

So, if you solve the equation v times x is 0 where v is a matrix given by writing the vectors,

this is an RN. If you look at that matrix vx is 0 where the columns of v are the vectors Vi

then, that the only solution is x is 0. And saying that it is linearly dependent means that there are

solutions which are not the 0 solution. So, this was linear dependence, and linear independence,

particularly in the context of RN. So, let us first talk about what is the

span of a set of vectors? So, span is like an expanse. So how big can you get

based on these vectors, that is the idea of span. So, the span of a set is, of vectors, is defined

as the set of all finite linear combinations of vectors of S, and it is denoted by span S.

So, explicitly what that means is, you take all linear combinations where your coefficients

run over all possible values in R, meaning each coefficient runs over all possible values in R,

and the span is that set. Now, what is hidden here is that,

the span of S is a vector space in its own rate, meaning it is what is called a subspace of V.

What that means is if you take a linear combination of elements within span S, then

they are still in span S. And if you take and zero is also that, in particular by taking all

the ai to be zero. So, here is an example. So, let S, be the set, just the singleton set 1 comma 0.

This is a vector in R2 and you look at the set S, which consists of this particular vector.

So, this is a subset of R2, and we can ask what is the span of S. So, the span of S is,

a times 1 comma 0, where a is in R which is the set of vectors, a comma 0. And we know

that this is exactly the x-axis in R2. So, the span of the vector 1 comma 0 is the x-axis.

So, let us do a couple of other examples. Let us look at the set 1 comma, the set consisting

of the singleton element 1 comma 1 in R2. So, you can see that the span of S is linear combinations

of 1 comma 1, which means a times 1 comma 1, which is all vectors of the form a comma a. So,

looking at this in R2, this means you get the line, y is equal to x? So,

the blue line is what is span of S? Again, you can see that, if you add vectors

on this line, you still remain on this line. So, let us see some more examples maybe in R3

this team. So, let us look at the set 1 0 0 and 0 1 0 in R3 and let us ask, what is the span of

this set? So, if you take linear combinations, you will get times 1 0 0, plus B times 0 1 0,

which is all vectors of the form a comma b comma 0. So, you can see, this is exactly the x,

y plane. So, span S in this case is the x, y plane, which is drawn here in the picture.

So, what are the spanning set for a vector space?

So, let V b a vector space. So, a spanning set is a set such that span of S is equal to V.

So, that means we are looking at vectors so that if you take linear combinations of those vectors,

you can produce any vector in V so, or the words, every vector of V is a linear combination

of vectors, belonging to the set S, that is what we mean by saying that span S is V.

So, let us look at the set 1 comma 0 and 0 comma 1 and R2,

then span of S is the entire R2, so this is a spanning set for R2. Similarly, if you take

1 0 0, 0 1 0, and 1 comma 2, so you add one more vector, clearly,

still the span is going to be the entire R2. And you can change your elements a little. So, if you

take the vectors 1 comma 1 and 0 comma 1 then the span of S is still R2. So, how do I see that?

Well, one way of seeing it is that from 1 comma 1 and 0 comma 1, you can produce 1 comma 0 by

taking 1 comma 1 minus 0 comma 1, so that will give you 1 comma 0. So, that means the

span contains 1 comma 0 and 0 comma 1. So, then if the span contains some vectors, then this span of

that set S is, also contains a span of the vectors that you have produced in that span. I will make

this more precise in the second. And let us look at an example in R3.

So, if you look at the set of vectors 1 0 0, 0 10 and 0 0 1, then span of S is R3. So,

let me quickly specify why that is the case. So let us maybe look first at 1 0 and 0 1,

so this first example, so if you take x comma y in R2 then you can write x comma y as

x times 1 comma 0 plus five times 0 comma 1. So, this means every vector in R2 can be written

as a linear combination of these first two vectors. So, that tells you that the span of

1 comma 0 and 0 comma 1 is indeed R2. The same idea can be used for this if you take x y z in R3

then x y z is x times 1 comma 0 comma 0, plus Y times 0 comma 1 comma 0

plus z times 0 comma 0 comma 1. So, this tells you that the span of these three vectors is the entire

R3. So, this is a spanning set for R3. Similarly, the first one was a spanning set for R2.

How about the second one? That is clear, I can still write x comma y as x times 1 comma 0 and

plus y times 0 comma 1. So, really this third vector here is kind of redundant. So,

the point here is, if your set S is a spanning set, then every superset is also spanning set.

So, that is that is what this example is trying to tell you. And then let us look at

the third example. So, for the third example what we want to say is the following.

So, you can of course write every x y directly as a linear combination of 1 comma 1 and 0 comma

1 which maybe I will do in a second. But what I really wanted to point out was that

1 comma 0 is 1 comma 1 minus 0 comma 1. So therefore, 1 comma 0 lies in the span of

this set 1 comma 1 and 0 comma 1 and certainly 1 comma 0 lies and certainly 0 comma 1 also

lies because it is one of those vectors in S. So, therefore, the set 1 comma 0 and 0 comma 1

belongs to contained in span of in this span. So, that implies span of this set is contained

in the span of this, which implies the span is R2. So, the main point here is the following,

so what I am trying to say is the following. If a set T is contained in a set S, then span of T

is contained in span of S. So, this is in fact what is a sub space.

And if it so happens that T is a spanning set, that means this is also spending set.

That is what we just said that if you have a superset, then we are good.

So, in particular one can also say the following that if T is contained in span of S,

which is what I am using here, then span of T is contained in span of S.

So, these are things that we will be doing as tutorial problems, so I will leave that for now.

So, let us continue this. So, I hope we have understood what is a spanning set,

so spanning set means a set such that every vector in V

can be written as a linear combination of this some particular vectors from S.

So how do I get a spanning set? So, the way to do this is, one way to do this is to

add or maybe the right word here is ‘append’ vectors to obtain a spanning set for R3, I will

use the word append instead of add, add may leave you confused by addition in the

vector space that is not what we mean. We mean append, meaning take more and more vectors.

So, we will try to build a spanning set for the vector space R3.

So, start with S0 to be the empty set, so this is the notation for the empty set,

a phi with a slightly tilted dashed on the middle.

So, then span of 0 is span of the empty set is 0 0 0. So,

there is something here already to point out, namely, that the span of the empty set is 0 0 0.

We can think of this in two ways. One is that this is a convention,

the other is that this is the empty linear combination. So, the empty linear combination

is always 0, that is again maybe a convention. So, from there it follows that 0 0 0 as the span

of the empty set. So, this, if you are not, if you find this uncomfortable then just take this as a

convention or a definition that span of the empty set is the 0 vector.

So, this is not the full vector space. The full vector space is R3,

so 0 0 0, is not the entire vector space R3. You append a vector outside span of S0 in R3,

say 0 to 1 to S0 and we will call this new set S1. So, S1 is S0 union

this new vector 0 to 1. So here, that is what I mean by append, you put this inside your,

you create a new set with this vector appended to your original set. And we are

going to keep continuing this process. This is how we will obtain a spanning set.

So, what is span of S1? So, span of S1 is exactly this line shown here. So, the blue arrow is 0

to 1, which is a vector that we had, and the span of S1 is the line containing that arrow.

So, now choose a vector outside span of S1. So, as we can see it is just the line,

so that means it is not the entire R3. So, we have vectors which are outside this line, so choose a

vector outside span of S1. So, for example, 2 2 0 appended to S1 and call that new set S2, so S2 is

S1 union with this appended vector 2 2 0. So, span of S2 is this plane that you get. So,

you can write down the equations for the plane, which we will be doing eventually, in this

in Maths 2. But if you can do it right away, maybe you have learned this already in Maths 1 or

maybe you have done it as part of some other course, please do go ahead and write down what the

equations for this plane are. In in any case, you get a plane and so this is not the entire R2, and

so we can choose a vector outside this plane. So, let us say choose a vector outside span of S2

say 0 0 5 and append it to S2 and we will call that new set S3. So, S3 is S2 union

0 0 5. So, you can see, we are increasing the size of the set that we have. So, you keep adding more

and more vectors, and hopefully as you keep doing this you will hit a spanning set.

So, now look at any arbitrary vector x y z in R3, and we can write it in the following way. So, this

is something you have to check x y z is y minus x by two times, 0 2 1 plus x by 2 times 2 2 0

plus x minus y plus 2z by 10 time 0 0 5. You might find this puzzling where did I suddenly

come up with these coefficients from, but you can certainly check that this is correct. So,

I will leave you to check it is correct. How to actually obtain this coefficient is

something we will see eventually or you can try to think of this as solving a system of linear

equations, we have done this before, and I will ask you to try to think about how to do this.

So, you can treat your coefficient as variables. We have done that in the previous video for linear

independence, and you can try and rate x y z as a linear combination of this.

So, what is the point? The point is,

span of S3 is the entire R3. How did I get that? Because x y z can be expressed as a

linear combination of these three vectors. So that means the span is the entire R3.

So, S3 is a spanning set for R3. So, what did we do? We started with the empty set

and then we kept adding vectors, but we added them in a particular way. We added them so that the

new vector that we added was not in the span of the vectors that we had so far. And in this way,

we created a spanning set. So, this is a general template to create a spanning set.

Let us do another example with this same idea. So, start with S naught with the empty set as before,

we know by convention or definition that the span of the empty set is 0 0 0

append any vector which is not in the span of S naught, let us say in this case, we choose 3 0 0

call this new set S1, so S1 is the set 3 0 0. And what is the span of S1? It is the x-axis. If

you take 3 0 0 and take it spanned that means you will get vectors of the form 3a comma 0 comma 0,

which is exactly what the vectors on the X-axis look like. So, the blue arrow is the vector 3 0

0. And the line of course is the X-axis. So, let us choose a vector, which is

outside the X-axis, let us say 2 2 1 and let us append it to S1 and call the new set

S2. So S2 is the set 3 0 0 and 2 2 1. Let us look at what is span of S2. Well, it is this plane,

this plane certainly passes to the X-axis. You can see it intersects the X-axis somewhere,

and it also contains the vector 2 2 1. So, I will, again, encourage you to check what

is the equation on this plane. In any case it is a plane, so it is not the entire R3.

So, you can choose a vector, which is outside this plane, which is the span of S2. So,

let us say, we choose the vector 1 3 3 and you appended to S2 and you call the new set S3.

So S3 is the set of vectors 3 0 0, 2 2 1 and the vector 1 3 3. And now once again, we will

sort of say that every vector x y z can be written as a linear combination of these three vectors,

and I will encourage you to work out what the coefficients are

by solving the corresponding system of linear equations.

If not, I have given the coefficients here and you can check that this indeed works out.

So, what is the point? The point is every vector in R3 can be written as a linear

combination of these three vectors. That means the span of S3 is the entire R3,

which means S3 is a spanning set for R3, which now brings us to this idea of a basis.

This is the main emphasis of this video, so what is a basis? So, basis B of a vector space V

is a linearly independent subset of V that spans V, so it has two properties. One, that it spans

V so it is a spanning set for V, and the other is that it is a linearly independent set.

Now, notice what is happening here. Span, if you want something to span V,

the idea is you keep adding elements. So, you add as many elements as you like, you keep adding

vectors and hopefully if you have enough vectors, then they will be able to span V.

Of course, we did this appending of vectors in a very particular way. We kept choosing vectors

which are outside the span of the previous few vectors that we have chosen. So, because we did

that, the new vector that we added was not a linear combination of the original vectors,

which meant it was linearly independent, the sets that we

keep obtaining are linearly independent sets. So, in fact, what we did in our previous

examples was a method to not just find a spanning set, but actually find a basis.

So, there are two contradict, I mean, there are two sort of opposite things happening here. One is

that you would like this set to be a spanning set, which means you would like it to be fairly large.

On the other hand, you would like it to linearly independent, which means

you do not want it to be too large. Afterall, remember that, for example,

if you have Rn then any set which has a cardinality more than N, so N plus 1 or more

is already linearly dependent. So, in RN, if you want to have a basis, you would want it to,

if you want it in particular to be linearly independent it can have size at most N. On

the other hand, for a spanning set you would have you would want it to be very large.

So, the basis is sort of an optimal middle. It is both linearly independent and spanning. So,

let us start with the most standard example. And in fact, this is called a standard basis.

So let ei belong Rn. We have seen this example by the way in our previous examples

of a spanning set. So, 1 0 and 0 1 is a spanning set for R2 it is actually a basis

and 1 0 0, 0 1 0 and 0 0 1 is a spanning set for R3 it is actually a basis and both of them

are what are called as standard basis and this is a more, I mean, in more generality

for any Rn, you can do the same thing. So ei be the vector with ith coordinate 1 and

all other coordinates 0, for example, e1 is the first coordinate is 1 and the other coordinates,

all the other coordinates are 0 so 1 0 0, all the way up to the nth coordinate, which is also 0.

So, let us look at the set. Let us say capital Epsilon, which is e1 e2 en this is a subset of Rn.

And the claim is that this is a basis for Rn. So, let us try to see why that is the case. If

you take – this is the same idea that we have done for in the case of R2 and R3 which where we

wrote it down. So, if you take anything of the form x1 x2 xn, I can write this as x1 times e1

plus x2 times e2 plus xn times en. So, it is a spanning set for Rn,

so what we have proved is that the span of this set Epsilon is Rn. On the other hand,

these are clearly linearly independent because if you take a linear combination, which is 0,

but let us expand out this vector in terms of coordinates.

So, what does the ith coordinator, what is jth coordinator? So jth coordinate of the

left-hand side

is aj, but for the right-hand side, the jth coordinate is 0. That means aj must be 0 and for

all j. That means it is linearly independent. So, therefore, this is linearly independent.

So that proves it is a basis. So, this is called the standard basis for Rn.

So, let us take a step back and ask, what have we done in this video. We have seen that

there is a notion of a spanning set, namely, a set of vectors so that every vector can be

written as a linear combination of these vectors, so that is a spanning set for the vector space V.

And then we have seen that there is a notion called a basis.

So, basis is a set of vectors, which is both linearly independent and spanning. Thank you.